Post-Training SNR-Gate Tabulation for Wireless Image Transmission: A Reproducible Study of WITT

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Abstract

We study post-training lookup-table deployment of the signal-to-noise ratio (SNR) gates in WITT, a transformer for wireless image transmission. Fourteen small conditioning networks depend only on scalar SNR, yet account for 4,150,272 learned parameters in our configuration. We replace these networks with 13-node FP32 tables and linear interpolation while retaining the image-dependent computation and channel-use ratio. For one locally trained checkpoint on CIFAR-10 with an additive white Gaussian noise (AWGN) channel, the complete serialized deployment file decreases from 54,987,545 to 38,644,489 bytes, a 29.72% reduction. Validation determines the table choice before a frozen evaluation on 10,000 test images, eight predeclared SNRs, and three paired noise realizations. The maximum signed decrease in peak signal-to-noise ratio (PSNR) is 0.000186 dB on these test conditions. This reproducible deployment study documents the storage–reconstruction tradeoff, a nearest-neighbor failure case, and the distinction between table knots and interpolated conditions.

1. Introduction

Learned joint source–channel coding (JSCC) maps images to channel symbols and reconstructs them after noisy transmission. Its encoder and decoder jointly determine reconstruction quality under a bandwidth and power budget [1]. SNR-adaptive systems condition this mapping on channel quality, allowing one trained model to operate at several SNRs [5, 6]. Deploying such a system also requires storing its conditioning networks, even when those networks receive a single scalar input.

WITT provides a concrete opportunity to examine this cost. In the author implementation used here, seven encoder gates and seven decoder gates each map scalar SNR to a 384-dimensional vector. These gates are independent of the transmitted image. Their outputs can therefore be evaluated before deployment and stored at selected SNR knots. Runtime interpolation then replaces the learned scalar-to-vector maps. The central question is whether removing their parameters can yield a useful reduction in complete model-file size while preserving reconstruction quality between the stored knots.

We investigate this question through an author-component reproduction and a post-training compilation procedure. A 13-node table removes 4,150,272 learned parameters and reduces the serialized model file by 29.72%. With the checkpoint and deployment choice fixed before testing, its maximum signed PSNR decrease is 0.000186 dB across eight predeclared CIFAR-10/AWGN test conditions. Five of those SNRs coincide with table knots; the remaining three exercise interpolation.

The study contributes a reproducible intervention on WITT's scalar-only gates and an evaluation that separates checkpoint selection, table selection, serialization checks, and final testing. A historical pilot compares linear interpolation with nearest-neighbor lookup and a shared-SNR computation control. Subsequent dense validation examines fractional SNRs on a stronger checkpoint. The resulting evidence characterizes a specific deployment tradeoff, with actual file sizes and paired reconstruction errors retained for independent inspection.

2. Related Work

Learned wireless image transmission. DeepJSCC learns a direct mapping between image pixels and complex channel inputs [1]. By optimizing reconstruction through a simulated channel, it couples source representation with resilience to transmission noise. Attention-based adaptive JSCC incorporates channel quality into feature processing [5]. WITT introduces a transformer architecture and SNR-dependent feature modulation for wireless image transmission [6]. SwinJSCC further develops Swin-transformer-based JSCC with channel and rate adaptation [7]. These studies motivate conditioning the encoder and decoder on communication constraints. Our experiment uses the original WITT components to isolate the storage cost of scalar SNR conditioning at a fixed channel-use ratio.

Lookup tables for neural deployment. Evaluating a learned function in advance and interpolating stored outputs

is an established deployment strategy. Jo and Kim transfer learned image mappings into lookup tables for practical single-image super-resolution [2]. The dimensionality of the table input is central to its storage cost: a multidimensional image neighborhood and a single channel-quality scalar lead to different table designs. Here, the tabulated input is only SNR. The table stores outputs of WITT's trained conditioning networks, while image-dependent feature transformations remain in place. The empirical contribution is the measured file-size and reconstruction tradeoff for this intervention, together with a test protocol that checks interpolation rather than only stored knots.

3. Model and Post-Training Compilation

3.1. Author-component reproduction

For an image $x \in [0,1]^{3\times32\times32}$, let f_θ and g_ϕ denote the WITT encoder and decoder. Both receive SNR s in decibels. The encoder produces real-valued latent features that the author channel implementation packs into complex symbols. After power normalization, the simulated channel adds independent real and imaginary Gaussian noise with standard deviation

$$\sigma(s) = \sqrt{1/(2\,10^{s/10})}. \tag{1}$$

The implementation restores the latent scale after transmission and passes the result to the decoder. Its normalization averages power over the batch; paired comparisons therefore preserve batch membership and order.

We use the author's CIFAR-10 configuration with latent width $C = 32$, two encoder stages of depths $[2, 4]$, embedding dimensions $[128, 256]$, and window size 2. The decoder reverses these stages. The channel bandwidth ratio (CBR) is $1/3$, defined as complex channel uses divided by real source samples. The original deployment has 13,715,056 learned parameters. The local training wrapper imports the author's model and channel components without modifying their source.

3.2. Scalar gate tabulation

For each of the 14 gates, let $h_j(s) \in \mathbb{R}^{384}$ be its trained scalar-to-vector map. Each map is a multilayer perceptron (MLP) with dimensions $1 \to 384 \to 384 \to 384$, hidden ReLU activations, and a final sigmoid. Including biases, it contains

$$(384 + 384) + 2(384^2 + 384) = 296{,}448 \tag{2}$$

parameters. The 14 maps together contain 4,150,272 parameters.

For K uniformly spaced knots $s_k \in [1, 13]$ dB, we evaluate $v_{j,k} = h_j(s_k)$ in FP32 after training. At runtime, for $s \in [s_k, s_{k+1}]$, we use

$$\widetilde{h}_j(s) = (1-\alpha)v_{j,k} + \alpha v_{j,k+1}, \qquad \alpha = \frac{s - s_k}{s_{k+1} - s_k}. \tag{3}$$

The main deployment uses $K = 13$, giving integer-dB knots. Both the knots and gate outputs are stored as buffers. Their total is $14 \times 13 \times (384 + 1) = 70{,}070$ FP32 values, or 280,280 raw bytes. The retained learned-parameter count is 9,564,784. All image-dependent feature transformations, the Swin backbone, the channel implementation, and CBR remain unchanged. Compilation requires no additional training.

Deployment	Parameters	File bytes
Original	13,715,056	54,987,545
Linear, 5 knots	9,564,784	38,472,457
Linear, 13 knots	9,564,784	38,644,489

Table 1. Current checkpoint (21,500 updates): complete inference-file sizes and learned-parameter counts. Table buffers are additional stored values, rather than learned parameters.

We measure storage from the complete serialized inference files, including metadata, buffers, and serialization overhead. These files exclude optimizer state. We separately check that 223 retained non-gate tensors are unchanged and compare a freshly compiled deployment with the same deployment loaded from its saved file. The recorded maximum output difference for this same-variant witness is zero. This checks serialization fidelity; the original-versus-table reconstruction difference is measured separately.

4. Experimental Protocol

Data and training. CIFAR-10 contains 50,000 training and 10,000 test images [3]. A fixed permutation with seed 20261107 partitions the training images into 45,000 training and 5,000 validation examples. Inputs are scaled to $[0, 1]$. We use Adam with learning rate 10^{-4}, batch size 128, and one training seed, 42. Training begins with 2,000 updates of the author's non-adaptive variant at 7 dB, followed by the adaptive model with uniformly sampled SNRs from $\{1, 4, 7, 10, 13\}$ dB. Adam is reset at this stage boundary. Training segments preserve optimizer, sampler, and random-number-generator states.

Every 500 updates, the adaptive model is evaluated at 1, 7, and 13 dB. Selection minimizes the arithmetic mean of the three MSEs, retaining the earlier checkpoint on ties. Among 41 eligible evaluations, the selected checkpoint occurs at 21,500 total updates. The training process later pauses at update 22,645; a validation performed only because of a cutoff is excluded from selection. All current

deployment variants use the selected checkpoint. Computation uses FP32, PyTorch 2.9.1 with CUDA 12.8, and an RTX 5070 Laptop GPU.

Validation before final testing. A historical pilot uses the 8,000-update checkpoint, 1,000 validation images, 13 fixed SNRs, and three paired channel-noise realizations. It compares 3-, 5-, 9-, and 13-node linear tables, a 5-node nearest-neighbor table, and a control that computes each original SNR MLP once per batch and broadcasts its output. The primary 13-node design is fixed after reviewing this pilot.

On the 21,500-update checkpoint, P002 compares the original model with 5- and 13-node linear tables using 1,000 validation images at 32 SNRs. These include every half-decibel point from 1 to 13 dB and seven additional fractional stress points. P003 compares the original and primary 13-node deployment on all 5,000 validation images at the eight SNRs subsequently used for testing. Each validation uses three paired noise realizations. Admission requires maximum signed PSNR loss at most 0.1 dB, at least 25% reduction in serialized bytes, and a successful serialization witness.

Both validations pass before the checkpoint, deployment files, and P004 test protocol are frozen. P004 evaluates all 10,000 official test images at $\{1, 1.5, 2.375, 3, 7, 12, 12.5, 13\}$ dB using three new paired noise realizations. Validation uses noise seed base 51001 and testing uses 91001. Testing makes no further selection of knots or model parameters. Its serialization witness uses a deterministic synthetic ramp, keeping that check separate from the official test images.

Metrics and conditional uncertainty. At each SNR, we clip reconstructed pixels to $[0, 1]$ and compute an MSE $e_{i,r}$ over the 3,072 scalar RGB channel values of image i in noise realization r. Reported PSNR is

$$P = -10 \log_{10} \left(\frac{1}{3N} \sum_{r=1}^{3} \sum_{i=1}^{N} e_{i,r} \right). \tag{4}$$

This is PSNR from pooled MSE, rather than the mean of per-image PSNRs. Signed loss is $\Delta = P_{\text{original}} - P_{\text{table}}$, so a negative value denotes a small improvement for the table.

For P002–P004, at each SNR, 2,000 paired bootstrap draws resample original batch IDs jointly across both methods and all three noise realizations. Each draw pools the recorded error sums and image counts, correctly weighting the shorter last batch. The full validation has 40 batch clusters (last batch: 8 images), and testing has 79 (last batch: 16). The 2.5th and 97.5th percentiles form pointwise descriptive intervals conditional on the checkpoint, fixed batch members, and recorded noises. The three noise realizations are repetitions of the channel simulation, rather than independent training seeds.

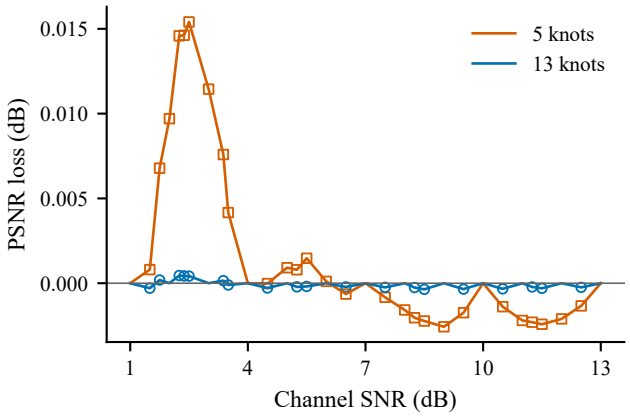


Figure 1. Dense validation (P002): original PSNR minus table PSNR at 32 fixed SNRs, with 1,000 images and three paired noise realizations per condition. Lines guide the eye between measured points; open markers identify non-knot conditions for each table. Five-node knots are 1, 4, 7, 10, and 13 dB; 13-node knots are every integer from 1 to 13 dB. Both variants use checkpoint 21,500.

5. Results

Deployment storage. Table 1 reports actual saved inference-file sizes. The main 13-node deployment occupies 38,644,489 bytes versus 54,987,545 for the original model, a 29.7214% reduction. Five nodes save a further 172,032 bytes, only a small additional reduction relative to the full original file. The measured storage improvement follows from removing the gate parameters while retaining compact FP32 lookup buffers.

Interpolation and table density. On dense P002 validation, the 13-node table's maximum signed loss is 0.000454 dB at 2.25 dB, whereas the 5-node table reaches 0.015405 dB at 2.5 dB (Fig. 1). Both pass the predeclared tolerance. The 13-node choice offers the smaller observed loss for a modest file-size increase. On all 5,000 validation images in P003, its maximum signed loss is 0.000264 dB at 2.375 dB.

Frozen official test. Table 2 gives absolute reconstruction PSNR and the paired difference on the frozen test grid. The maximum signed decrease is 0.000186488 dB at 2.375 dB, with descriptive interval $[-0.0000406, 0.0004124]$ dB. The maximum absolute change is 0.000337846 dB at 1.5 dB; its negative signed loss indicates a small improvement. These are different summaries of the same paired results.

Five test conditions are integer-dB table knots and show differences below 3×10^{-8} dB in magnitude. The three fractional conditions, 1.5, 2.375, and 12.5 dB, exercise interpo-

SNR (dB)	Original (dB)	13 knots (dB)	Loss (mdB)	95% interval (mdB)
1	27.09260	27.09260	0.000	[0.000, 0.000]
1.5	27.53395	27.53429	-0.338	[-0.479, -0.188]
2.375	28.27304	28.27285	+0.186	[-0.041, 0.412]
3	28.78591	28.78591	0.000	[0.000, 0.000]
7	31.84959	31.84959	0.000	[0.000, 0.000]
12	34.87276	34.87276	0.000	[0.000, 0.000]
12.5	35.11328	35.11354	-0.258	[-0.269, -0.246]
13	35.33646	35.33646	0.000	[0.000, 0.000]

Table 2. P004 official test: 10,000 images and three paired noises at each SNR. PSNR is computed from pooled MSE. Loss is original minus 13-node PSNR, in millidecibels (mdB; $1\,\mathrm{mdB} = 10^{-3}$ dB). Intervals are conditional, pointwise, and descriptive (Sec. 4). Values, including near-zero knot differences, are rounded for display.

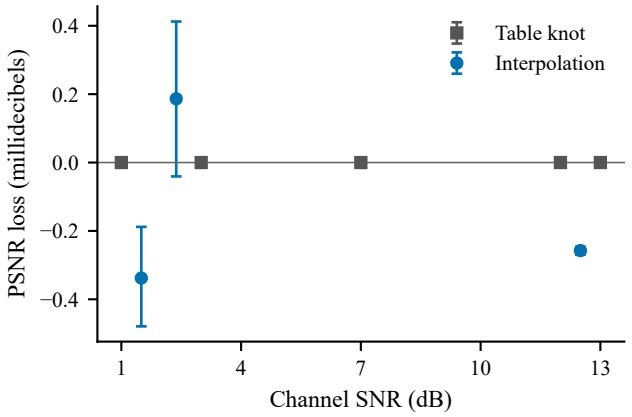


Figure 2. Frozen P004 test differences and conditional pointwise 95% descriptive intervals from 2,000 paired batch-cluster bootstrap draws. Each condition uses 10,000 images, three noises, and 79 batch clusters. Squares mark table knots; circles mark interpolation. Positive loss means lower PSNR for the table.

lation. Figure 2 displays these differences in millidecibels so that their scale and conditional intervals remain visible. Both positive and negative differences occur, as expected for a small perturbation of a learned model rather than a monotonic quality improvement.

Historical controls and failure case. Table 3 retains all pilot variants on their original 8,000-update checkpoint and 13-point validation grid. Linear interpolation with three knots has maximum loss 0.093756 dB; nearest-neighbor lookup with five knots has larger loss, 0.129533 dB, despite the latter's denser table. Thus, lookup resolution and interpolation rule both matter. The shared-SNR control preserves the original MLP parameters and shows a near-zero reconstruction difference, separating repeated scalar computation from parameter removal. The table values describe this historical checkpoint and grid; the dense current-checkpoint results above are a distinct experiment.

Method	File bytes	Max. loss (dB)
Original	54,987,481	0.000000
Shared-SNR MLP	54,989,849	0.000000
Linear, 3 knots	38,429,449	0.093756
Linear, 5 knots	38,472,457	0.009071
Linear, 9 knots	38,558,473	0.000389
Linear, 13 knots	38,644,489	0.000258
Nearest, 5 knots	38,472,457	0.129533

Table 3. Historical P001 pilot: 8,000-update checkpoint, 1,000 validation images, 13 SNR conditions, three paired noises. Maximum signed loss is taken over the full recorded grid. This checkpoint is separate from Tables 1 and 2.

6. Discussion and Conclusion

Tabulating WITT's scalar SNR gates removes a substantial parameter block while preserving its image-dependent processing. For the selected CIFAR-10/AWGN checkpoint, a 13-node deployment reduces complete serialized storage by 29.72%, with maximum signed PSNR decrease 0.000186 dB on the frozen eight-point official test. Dense validation and a nearest-neighbor comparison show why evaluating off-knot conditions is essential when compiling a conditional neural network into tables.

Limitations and next steps. The evidence covers one architecture, one training trajectory, CIFAR-10 images, simulated AWGN, and CBR $1/3$. Modern libraries, a validation holdout, and a finite update budget differ from the author's training protocol; numerical parity with the published WITT results and convergence remain unestablished. The two checkpoints share one training trajectory. The 13 knots cover 1–13 dB; the implementation clamps outside this range, and domain-exterior behavior is untested. Eight test points and conditional intervals support pointwise conclusions rather than a continuous-SNR guarantee. PSNR measures image reconstruction, while downstream semantic-task utility, real radio or satellite links, and other channel models require separate evaluation. Fixed-order pilot timings vary and provide no established latency advantage. Further work should examine additional training seeds and architectures, rate adaptation, and device-level storage and latency together.

References

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A. Reproducibility and AI-Assisted Workflow

Source and execution. The author WITT implementation is pinned to commit `2460db5` at https://github.com/KeYang8/WITT. Evaluation records identify the full commit and original source status; deployment files also retain source metadata. The training, compilation, evaluation, and analysis entry points are `train_witt.py`, `prepare_gate_tables.py`, `evaluate_gate_tables_paired.py`, and `analyze_paired_gate_results.py`, respectively. Plot-generation scripts read saved JSON results and require no model inference. The local package contains configuration files, raw metrics, figure sources, environment specifications, and a reproduction guide. Public hosting and author details remain pending for this course draft.

Saved evidence. The selected checkpoint is identified by `R003-checkpoint-selection.json`. P002's preparation record supplies parameter counts and complete file sizes. P002 and P003 retain validation metrics; P004 retains per-method, per-noise, per-batch error sums and counts on the official test images. The test-freeze record precedes testing. Source snapshots accompanying P002–P004 preserve the evaluator and protocol. Original dataset pixels are separate from these source snapshots.

Actual AI use. The project follows the ARIS skill-based research workflow [4], with Codex as the executing AI. The workflow covers baseline selection, author-code inspection, resumable local training, candidate generation, critical review, method refinement, gated evaluation, and evidence review. Initial work executed several subflows; a user correction led to explicit main-pipeline tracking and the later review–revision–evaluation cycle. Reviewer identities and raw replies are recorded. The reviewers belong to the same model family as the executor, so their assessments remain provisional.

One critical review corrected the three-node pilot's full-grid worst loss from a narrower-subset value to 0.093756 dB, identified missing fractional SNR conditions, and motivated the stronger validation protocol. Code review then required a synthetic serialization witness for testing and verification of actual deployment sizes and checkpoint identity before opening test images. Both fixes preceded P004. A final joint evidence review checked experimental integrity and bounded claims; it was one review session serving several checklists, rather than multiple independent reviews. The resulting artifact is a CVPR-formatted course study. Student-authored reflections and publication decisions are recorded separately from measured scientific results.